Project template

https://github.com/AdrianUrbanski/Neural-Networks-Project-Template
This repository contains a project template for the course. Its usage is not mandatory, but it might make managing the project easier. It provides a clean, working starting point for training neural networks, demonstrating how three tools fit together in a typical deep learning workflow:

- PyTorch Lightning — structures training code around two abstractions: LightningModule (model architecture, loss, optimizer, train/val steps) and LightningDataModule (dataset, splits, transforms, dataloaders). The entry point scripts/train_model.py is fully model-agnostic — adding a new model or dataset requires no changes to the training script.
- Fiddle (Google) — a Python-native configuration system. Each experiment is defined as a config file that returns a fully specified, serializable config object. Configs can inherit from a base and override individual values, making hyperparameter sweeps (e.g. learning rate comparisons) straightforward and reproducible.
- Weights & Biases (W&B) — experiment tracking. Every run logs metrics, hyperparameters, and the exact Fiddle config used, so results are reproducible and comparable across runs.

The included models (CNNs and MLPs on MNIST and UTKFace) are intentionally simple — their purpose is to illustrate the abstractions, not to be state-of-the-art. The README contains a more detailed description of the project structure and installation instructions. If you want to understand how the pieces connect, scripts/train_model.py is a good starting point — it is well-commented and walks through the full training flow.